W1 DESIGN OF EXPERIMENTS

Aim: decide whether data can support **causal** claim or only assoc. Units, treatment/exposure, response, comparison group, allocation, measurement.

Causation needs: comparable groups + controlled exposure + plausible mechanism + no major bias/confounding.

Experiment: investigator assigns treatment. Best: random allocation, placebo/control, double blind, contemporaneous control.
Observational: no assignment; can estimate assoc, not alone prove cause. Useful when randomisation unethical/impossible.
Confounder: variable related to exposure and response, not on causal pathway; mixes effects. Eg app use vs final mark: prior WAM affects both.
Check causal story: 1 treatment before outcome; 2 identify comparison; 3 list confounders; 4 ask random/blocked/blind? 5 decide assoc vs cause.
Bias map: selection bias=who enters; consent bias=volunteers differ; observer bias=measurer expectations; survivor/dropout bias=missing worst cases; recall bias=memory differs.
Placebo effect: response to belief of treatment. **Double blind**: subject + investigator unaware. **Historical control**: time may confound.

Simpson: subgroup trends can reverse when pooled if group sizes differ by confounder. Always compare within levels of confounder first.

Claim	Need	Weakness
Cause	random assign	noncompliance
Assoc	representative obs	confounding
Prediction	holdout data	leakage

Exam verbs: identify=state variable; explain=mechanism; critique=why inference fails; improve=design fix.

W2 GRAPHICAL SUMMARIES

Variable types: categorical nominal/ordinal; quantitative discrete/continuous. Choose plot by variable pairing.

Data	Plot	Reads
1 cat	bar	counts/props
1 quant	hist/box	shape/centre/spread
quant by cat	comp box	group med/IQR/outliers
2 quant	scatter	trend/form/strength/outliers
2 cat	mosaic/cluster bar	conditional props

Graph read: context -> axes/units -> overall pattern -> exceptions -> relevant comparison -> cautious conclusion.
Histogram: bin width changes impression; use for shape. Left-closed right-open $[a, b)$ common. Frequency counts in interval; density area if scaled.
Boxplot: median, Q1,Q3,IQR, whiskers to non-outliers, outlier if $< Q1-1.5IQR$ or $> Q3+1.5IQR$. Cannot read mean/SD exactly.
Scatter: direction (+/-), form (linear/curved), strength, outliers/clusters.
Correlation only linear.

Do not infer causation from plot; ask design. Overplotting: use alpha, jitter, faceting; alpha does not fit model.

ggplot: use point transparency, colour/facet by group, and clear axes; mapping inside aes varies by data.
Bad graphs: truncated axes exaggerates; dual axes confuse; pie poor for many cats; counts hide population size; stacked bars hide conditional denominator.

W3 NUMERICAL SUMMARIES

Centre: mean $\bar{x}=\sum x_i/n$ sensitive to outliers; median robust. Use median for skew/outliers.
Spread: $s=\sqrt{\sum(x_i-\bar{x})^2/(n-1)}$ sample SD; IQR=$Q3-Q1$ robust; range=max-min.

Shift/scale: if $Y=a+bX$, mean$_Y=a+b$mean$_X$, SD$_Y=|b|$SD$_X$, IQR$_Y=|b|$IQR$_X$; adding constant changes centre not spread.

Percentile: pth percentile has about $p\%$ below. Five-number summary=min,Q1,med,Q3,max.
Skew: mean > median often right-skew; mean < median left-skew. Outliers pull mean and SD more than median/IQR.
Compare groups: centre difference + spread + overlap + skew/outliers + sample size + practical context.
Correlation: r in $[-1, 1]$, unitless, unaffected by adding constants or multiplying by positive constant; sign flips if one variable multiplied by negative.

$r=0$ means no linear assoc, not no relationship. $r=.95$ does not mean 95% points on a line.

Transform	Mean	SD
$x+c$	$+c$	same
cx	$\times c$	$\times\lvert c\rvert$
$a+bx$	$a+b\bar{x}$	$\lvert b\rvert s$

Recall trap: every value +100 -> mean/median +100; IQR/SD/correlation unchanged.

W4 NORMAL MODEL

Normal: $X\sim N(\mu, \sigma^2)$, bell-shaped, symmetric. Standardise $Z=(X-\mu)/\sigma$.

$P(a\le X\le b)=\Phi((b-\mu)/\sigma)-\Phi((a-\mu)/\sigma)$; R: 'pnorm(b,mu,sd)-pnorm(a,mu,sd)'.

68-95-99.7: within 1,2,3 SD approx .68,.95,.997. Symmetry: $P(Z>z)=1-\Phi(z)=\Phi(-z)$.
qnorm: inverse probability/cutoff. 'qnorm(.975,mu,sd)' = upper 2.5% cutoff for two-sided 95%.
Measurement model: reading = exact + bias + random error. Repeated avg reduces random error SE $=\sigma/\sqrt{n}$ but not bias.

Averaging fixes noise, not systematic bias. Calibration fixes bias. More repeated readings shrink random SD of mean by $\sqrt{n}$.

Approx normal?: plausible for many biological measurement errors and sample averages; poor for bounded, highly skewed, tiny n, mixture distributions.

Task	R	Avoid
prob below	pnorm(x,mu,sd)	qnorm
prob above	1-pnorm(x,mu,sd)	sd^2 as sd
cutoff	qnorm(p,mu,sd)	pnorm

Unit conversion: if lb=2.205kg, both mean and SD multiply by 2.205; probabilities may compute on either scale after converting cutoff.

W5 LINEAR MODEL

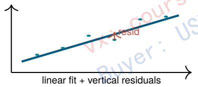
SLR: $\hat{y}=a+bx$, slope b expected change in y per 1-unit x; intercept at $x=0$ may be meaningless.

Prediction at x_0: plug into fitted line. Residual $e=y-\hat{y}$. RMS error $\approx$ typical vertical prediction error in y units.

Least squares: line minimises sum squared residuals; residuals average 0 when intercept included.
R output: coefficients table gives intercept and slope. Formula 'lm(y x)' predicts $\hat{y}=\hat{\beta}_0+\hat{\beta}_1 x$.
Model checks: scatter roughly linear; residuals random around 0; no U-shape/funnel/leverage; avoid extrapolation outside observed x.

R^2 is proportion of response variation explained in sample; high train R^2 does not guarantee low test error.

Correlation-slope link: $b=r(s_y/s_x)$. Correlation has no units; slope has units y/x. Slope comparisons across groups do not imply correlation comparison.



linear fit + vertical residuals

W6 UNDERSTANDING CHANCE

Chance: long-run freq, not guarantee in short run. Independence: knowing A gives no info about B; mutually exclusive events usually not independent unless one has prob 0.

Addition: $P(A\cup B)=P(A)+P(B)-P(A\cap B)$. Multiplication for independent: $P(A\cap B)=P(A)P(B)$.

Counting: with replacement independent; without replacement dependent but symmetry can simplify (5th card queen prob $1/52$).
Binomial: fixed n, independent trials, same p, count successes. $P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$, $E=np$, SD=$\sqrt{np(1-p)}$.
Box model: tickets represent possible outcomes; draws model randomness. Sum of draws: EV=$n\mu_{box}$, SE=$\sqrt{n}\,SD_{box}$. Avg: EV=μ_{box}, SE=$SD_{box}/\sqrt{n}$.

For 0/1 box with proportion p: mean=p, SD=$\sqrt{p(1-p)}$ approx; under $p=.5$, SD=.5.

Build box: define one draw outcome -> tickets -> number of draws -> statistic sum/avg/count/prop -> EV/SE -> compare observed.
Trap: SE for sum grows as $\sqrt{n}$; SE for average/proportion shrinks as $1/\sqrt{n}$.

W7 PROJECT WEEK

EDA cycle: question -> data provenance -> clean/types -> summaries/plots -> anomalies -> model/test if justified -> communicate limitations.

Client report = answer the practical question, not list every plot. Use context, units, effect size, uncertainty, and design caveats.

Data quality: missingness, impossible values, duplicates, coding choices, outliers, inconsistent units, sampling frame, privacy/re-identification.
Reproducible workflow: import -> wrangle -> visualise -> summarise -> model/test -> interpret -> cite sources; scripts + seeds + clear variable definitions.
Ethics: privacy, consent, harm, fairness, transparency. Removing names not enough if rare combinations re-identify people.
AI use: allowed in open tasks only with acknowledgement; never fabricate data/output; verify code and claims; secure exams prohibit AI.

Problem	Symptom	Fix
leakage	test info in train	split first
bad denominator	wrong percent	state base
confounding	causal overclaim	stratify/design
p-hacking	many unplanned tests	pre-specify

Presentation: claim -> evidence -> caveat -> action. If plot and number conflict, inspect definitions and subsets before choosing.

W8 BOX MODEL/CLT

CLT: for large n, distribution of sums/averages is approx normal even if box not normal, if independent draws and no single huge-ticket domination.

Observed statistic -> expected under model -> SE -> z=(obs-EV)/SE -> p-value/tail area.

Sample avg: $\bar{X}\approx N(\mu, \sigma^2/n)$ for large n. **Sample sum**: $S\approx N(n\mu, n\sigma^2)$.
Finite population: without replacement from small population lowers SE via finite population correction; DATA1001 often uses with-replacement/simple approx unless specified.
Simulation: repeat random experiment many times; empirical histogram approximates sampling distribution. More reps reduce simulation noise, not model bias.

A 95% normal approx interval for statistic: estimate $\pm 2SE$ rough; exact interpretation depends on statistic and sampling.

box tickets -> n draws -> sum/avg -> normal approx

Diagnostic: if n small and box very skewed/binary rare event, prefer exact binomial/simulation over normal approx.

W9 SAMPLE SURVEYS

Population: group of interest. Sample: observed subset. Parameter fixed unknown; statistic random.
Sampling: SRS gives every unit equal chance; stratified samples within key groups; cluster samples groups; convenience/volunteer samples risk bias.

Bias does not shrink with larger n; random error shrinks as $1/\sqrt{n}$. A huge biased sample can be precisely wrong.

Sample proportion: $\hat{p}$ estimates p; SE approx $\sqrt{p(1-p)/n}$ or plug-in $\sqrt{\hat{p}(1-\hat{p})/n}$. Max SE at $p=.5$.
Survey critique: sampling frame -> selection method -> nonresponse -> wording/order -> mode -> weighting -> uncertainty.
Question wording: leading/loaded/double-barrelled questions shift responses. Eg "stop wasting money..." invalid neutral wording.
Margin of error: rough 95% MOE $\approx 2SE$. For percentages, convert proportions carefully.

Error	From	Fixed by
random	chance sample mix	larger n
selection	bad frame	better design
nonresponse	who replies	follow-up/weights
measurement	bad question/tool	redesign

External validity: even perfect random sample only generalises to the sampled population/frame.

W10 HYPOTHESIS TESTING

Logic: assume H_0, compute how surprising data are; p-value = probability of statistic at least as extreme as observed under H_0.

Small p -> evidence against H_0; large p -> data consistent with H_0, not proof H_0 true.

Test recipe: 1 parameter; 2 H_0, H_1 before seeing data; 3 test statistic; 4 reference distribution/simulation; 5 p-value; 6 conclusion in context.
Alternatives: two-sided $\neq$; one-sided $>$ or $<$. Direction must be pre-specified; if observed direction opposite one-sided alternative, p large.
Type I: reject true H_0, prob α. **Type II**: fail reject false H_0. Power=$1-\beta$.

Multiple tests inflate false positives: expected false rejections $m\alpha$ if all null true.

Test statistic: signal/noise, e.g. $(\hat{\theta}-\theta_0)/SE$. Larger absolute value often stronger evidence.
P-value myths: not $P(H_0 true)$; not effect size; not probability result due to chance; not reproducibility guarantee.

W11 TESTS FOR A MEAN

One-sample mean: $H_0:\mu=\mu_0$; statistic $t=(\bar{x}-\mu_0)/(s/\sqrt{n})$, df=$n-1$ if normal/t approx appropriate.

R 't.test(x,mu=mu0)' default two-sided; CI containing μ_0 corresponds fail reject at matching two-sided α.

Assumptions: independent sample, quantitative response, no extreme skew/outliers for small n; larger n uses CLT.
Interpret CI: plausible values for population mean under method; not interval containing 95% of observations.
One-sided from two-sided: if symmetric and effect in predicted direction, one-sided p approx half two-sided p; if wrong direction, p approx $1-$half.

Effect size matters: statistical significance can be practically tiny with large n; non-significance may be low power.

Output	Read	Conclusion
p=.024	$<.05$	reject at 5%
CI [48.5,52.7]	contains 50	fail reject 50
t=-2.1	below null	sample mean lower

No calculators?: focus on form, direction, compare p to alpha, and contextual wording.

W12 TESTS FOR RELATIONSHIP

Proportion test: binary outcome, $H_0: p=p_0$; under $p_0=.5$, SE=$.5/\sqrt{n}$ for $\hat{p}$. Taste-test "no preference" = $p=.5$.
Chi-square GOF: compare observed counts to expected counts. $\chi^2=\sum(O-E)^2/E$, df=categories-1-adjusted constraints; large values reject.
Chi-square independence: two categorical variables; expected E_{ij} = row total × col total/n; df=$(r-1)(c-1)$.
Regression relationship: test slope/correlation vs 0; interpret slope, p-value, CI, residual diagnostics. Association not causation unless design supports.

For relationships: state variables, direction, strength/effect, uncertainty, design limitation.

Choose test: binary one sample -> prop/binomial; one quantitative mean -> t; many category counts -> chi-square; two quantitative relationship -> regression/correlation.
Common final traps: wrong denominator for percentages; using qnorm for probability; accepting null; ignoring confounder; comparing training R^2 to test error; assuming larger samples removes bias.

Question	Method	Key output
mean=?	t.test	t,df,p,CI
prop=.5	prop/binom/box	z,p
fair die	chi-square GOF	χ^2,df
x-y linear	lm/cor	slope,r,resid

MICRO EXAMPLES

1) Box $\{0, 1, 1, 3, 5\}$, 100 draws sum: mean ticket 2, EV=200, SE=$10SD(box)$.
2) $X\sim N(12, 3^2)$, $P(10\le X\le 14)$='pnorm(14,12,3)-pnorm(10,12,3)'.
3) 'lm(mark sleep)' coef $(48, 3.2)$: predicted mark at sleep=7 is 70.4; do not call causal if sleep not randomised.
4) p=.78: little evidence against H_0; never say 78% chance null true.

R OUTPUT DECODER

R line	Means	Exam response
'summary(x)'	Min,Q1,Med,Mean,Q3,Max	shape/centre/outliers
'sd(x)'	sample SD	typical distance
'cor(x,y)'	linear assoc	sign/strength only
'lm(y x)'	fit line	predict/read slope
't.test'	mean test + CI	compare p/CI
'chisq.test'	count table test	large stat rejects

Reading code: identify data object, variable names, grouping formula 'y group', function purpose, arguments, output scale.
Formula syntax: 'response predictor'; in boxplot/lm/t.test, left side is quantitative response, right side group/predictor.
Common R traps: 'pnorm' gives area, 'qnorm' gives cutoff; 'sd=3' not variance 9; 'alpha' in ggplot is transparency; 'df' for t is not sample size.

When R output and story disagree, trust definitions/variables first: units, filtering, missing data, denominator, and whether values are proportions or percentages.

TEST CHOICE MAP

Data	Parameter	Null	Method
one mean	μ	$\mu=\mu_0$	t/box
paired diff	μ_d	$\mu_d=0$	t on diff
one prop	p	$p=p_0$	binom/box
counts k cats	$p_1..p_k$	specified	chi GOF
two cats	assoc	indep	chi indep
two quant	slope/r	0	lm/cor

Conclusion template: Since p [</>] α, [reject/fail reject] H_0. The data [provide/do not provide] evidence that [contextual alternative].

Fail to reject = insufficient evidence; never "accept", "prove equal", or "no effect exists".

Tail choice: "different/changed/associated" -> two-sided; "greater/increase/higher" -> upper; "less/decrease/lower" -> lower. Must choose before data.
Test stat sign: positive if observed estimate above null; negative if below. Two-sided uses magnitude; one-sided cares direction.
Critical value: reject if statistic beyond critical region; p-value method equivalent when same α and same tail.

STUDY DESIGN FIXES

Weak study	Main risk	Better design
self-select app	confounding	random assign app
historical control	time trend	concurrent control
open label	placebo/observer	double blind
convenience poll	selection	SRS/strata
high dropout	attrition	intention-to-treat
many outcomes	p-hack	pre-register

Blocking/stratifying: compare or randomise within known confounder groups to improve balance; still only controls measured factors.
Randomisation: balances known/unknown confounders on average; individual realised samples can still be imbalanced, esp small n.

A representative sample helps estimate population; random assignment helps causal inference. They solve different problems.

Causal critique paragraph: "The groups differ in ..., which may affect both ... and .... Therefore the observed difference could be due to ..., not the treatment."
Reverse causation: high marks may cause app engagement; health may affect exercise; disease symptoms may cause medication use.

GRAPH/NUMBER PITFALLS

Anscombe quartet: same mean/SD/correlation/regression can hide different shapes; always plot.
Ecological fallacy: group-level association may not hold for individuals.
Aggregation bias: combined data hides subgroup structure.

Percent comparison needs same denominator. "More students" counts can conflict with "higher rate" percentages.

Evidence	Stronger read	Caution
med far below mean	right skew	outlier?
wide IQR	variable middle 50%	units matter
high r	linear trend	outliers/curve
U residuals	missing curvature	transform/add term
funnel residuals	nonconstant var	predictions uneven

Outlier handling: do not delete just because inconvenient. Ask measurement error? valid rare case? analysis sensitivity? effect on mean/slope?
Standardisation: $z=(x-\bar{x})/s$ converts to SD units; useful across scales; does not make non-normal data normal.

BOX MODEL DETAILS

Tickets: encode null distribution, not observed sample unless using bootstrap-like resampling. For a fair die, box $\{1, 2, 3, 4, 5, 6\}$; for no preference, box $\{0, 1\}$ with equal tickets.

Observed count of n: observed prop $\hat{p}=k/n$; null EV count np_0; null SE count $\sqrt{n}\sqrt{p_0(1-p_0)}$.

Simulation p: simulate many stats under H_0; p = fraction at least as extreme as observed. Include both tails for two-sided.
Exact vs approx: binomial exact for small binary; normal/CLT for large; chi-square needs expected counts not too tiny.
Continuity intuition: count is discrete; normal is continuous, so approximations can be rough near boundaries or small counts.
SE is about repeated sampling variability, not variability among individual observations.

REGRESSION EXTRAS

Leverage: unusual x can strongly affect line. **Influential**: removing point changes fit a lot.
Prediction vs explanation: a model may predict well without causal explanation; causal explanation needs design/theory.

Do not extrapolate: line fit for observed x range may fail outside it. Intercept often meaningless if $x=0$ outside data.

Quantity	Units	Meaning
slope	y per x	change per 1 x
intercept	y	fit at x=0
residual	y	actual-pred
RMS err	y	typical error
r	none	linear strength
R^2	percent	in-sample variation

Residual answer: describe pattern around 0, spread, outliers, and implication for linearity/constant variance; avoid "model perfect" unless residuals all exactly 0.

INTERPRETATION SENTENCES

Slope: For each additional [unit x], predicted [y] changes by [slope] [y units], on average in this fitted data.
Correlation: There is a [weak/moderate/strong] [positive/negative] linear association between X and Y.
CI: We are 95% confident the method captured the population mean difference between ... and ....
P-value: If H_0 were true, a result this extreme or more would occur with probability p.
SE: Across repeated samples of this size, the statistic would typically vary from its expectation by about SE.
Bias: The sampling/measurement method systematically pushes estimates away from truth in one direction; more data from same flawed process does not fix it.

FAST NUMERIC CHECKS

Situation	Quick calc	Meaning
$n=100, p=.5$	SE prop=.05	10% diff is 2 SE
$n=400, \sigma=2.5$	SE avg=.125	avg very stable
add 4 to all	SD same	spread unchanged
multiply by -2	r sign flips	SD doubles
p=.032 two-sided	one-sided=.016	if right direction
20 tests .05	EV false=1	not guaranteed

Rounding: keep enough digits in intermediate steps; final context often 2-3 s.f. or as given by R.

EXAM STRATEGY

MCQ elimination: remove causation overclaims, p-value myths, wrong function ('qnorm' vs 'pnorm'), wrong denominator, spread changed by shift, impossible statistic.
Long question: answer s... then interpret; every conclusion ...

For every statistical claim, ... easured how? sampled from whom? ran... ertainty?

High-yield past-paper themes: ... ean/SD/IQR shifts; normal R code; reg... SE; survey bias; p-value wording; chi-...
Last check: units, directio... nd whether the question asks evidenc... iption.

COMMON LONG-Q PATTE...

Data story critique: ident... explanatory variables, population, desi... hether graph/summary answers question.
Simpson answer: calculate/compare subgroup rates, then combined rates; name lurking variable; explain unequal subgroup weights; conclude combined result is misleading for causal/group comparison.
Summary answer: quote relevant median/mean/IQR/outliers; compare groups using both centre and spread; mention skew if mean far from median; avoid overclaiming from one summary.
Normal answer: define variable and parameters, standardise cutoff(s), use pnorm/empirical rule, interpret probability in context.
Regression answer: write fitted equation, compute prediction, residual if actual given, assess residual plot, discuss extrapolation/causation.
Box/test answer: state H_0/H_1, build box, compute EV/SE/test stat, compare p/critical, conclude in words.

MARKING LANGUAGE

Weak phrase	Better phrase
proves	provides evidence
no relationship	no clear linear assoc
accepts null	fails to reject
average person	sample mean/median
random data	random sample/draws
bias random	bias systematic
normal because n big	avg approx normal by CLT

Always separate **sample statement** from **population statement**. "In this sample..." is descriptive; "in the population..." needs sampling model.

Practical significance: ask whether magnitude matters in real context, not just whether p is below .05.
Uncertainty: smaller SE/CI width means more precision, not less bias; a biased estimator can have tiny SE.

FORMULA BANK

Object	Formula
sample mean	$\bar{x}=\frac{1}{n}\sum x_i$
sample SD	$s=\sqrt{\frac{1}{n-1}\sum(x_i-\bar{x})^2}$
z-score	$(x-\mu)/\sigma$
t-stat	$(\bar{x}-\mu_0)/(s/\sqrt{n})$
prop SE null	$\sqrt{p_0(1-p_0)/n}$
prop SE plug-in	$\sqrt{\hat{p}(1-\hat{p})/n}$
binomial	$\binom{n}{k}p^k(1-p)^{n-k}$
chi-square	$\sum(O-E)^2/E$
expected table	row×col$/n$
residual	$y-\hat{y}$
RMS error	$\sqrt{\text{mean}(e_i^2)}$

Scale sanity: probabilities in $[0, 1]$; SD/SE nonnegative; correlation in $[-1, 1]$; chi-square nonnegative; p-value in $[0, 1]$.

EDGE CASES

SD zero: all observations identical; correlation undefined if either variable has zero SD.
Median vs mean: equal mean/median does not guarantee symmetry; symmetric data often mean=median but reverse not always true.
Outlier threshold: lower fence can be negative even if all observations positive; fence is a rule, not an observed value.
Mutually exclusive: $A\cap B=\emptyset$; if both have positive prob, observing A makes B impossible, so not independent.
Correlation after swap: corr$(B, A-100)$ = corr(A, B); subtracting constants no effect, swapping no effect.
Negative scale: corr$(-2X+5, Z)$ = $-$corr(X, Z); SD multiplies by 2; mean becomes $-2\bar{X}+5$.

Impossible answers often violate range/units: negative SD, p-value >1, median below min, $R^2<0$ for standard in-sample lm with intercept.

PRIVACY/ETHICS

De-identification: removing name/SID may fail if rare combinations identify people. Aggregate, suppress, perturb, or restrict access.
Sensitive variables: health, disability, postcode, age, small program, ethnicity. Risk rises when combined.
Ethics answer: purpose limitation, consent/notice, minimisation, secure storage, fairness, possible harm, transparent reporting.

A technically valid analysis can still be ethically poor if collection/use harms or misleads the population studied.

MINI DRILLS

Given $\bar{x}=58$, median=42, max=140: likely right-skew; mean pulled up.
Given t=1.2, p=.24, CI [48.5,52.7] for $\mu=50$: fail reject; 50 plausible.
Given $\chi^2=11.8$, df=3, 99% cutoff 11.34: reject at 1% because large chi-square.
Given app users higher marks and higher prior WAM: prior WAM confounds; cannot infer app caused marks.
Given 52/80 prefer A under $p=.5$: $\hat{p}=.65$, SE=$.5/\sqrt{80}$, statistic $(.65-.5)/SE$.
Given residuals positive-low, negative-middle, positive-high: curved relation missing.
Given 92% response to leading survey: high response does not fix loaded wording.
Given twenty true null tests at $\alpha=.05$: expected false positives 1, actual count random.